Sriram Vijendran

Ames, Iowa | vijendran.sriram@gmail.com | github.com/sriram98v | linkedin.com/in/sriram-vijendran-02595512a | sriram98v.github.io

Education

Iowa State University — Ames, IA
Ph.D. in Computer Science — *Jan 2021 — Present*

SRM Institute of Science and Technology — Chennai, India
Bachelor of Technology in Electronics Engineering — *Aug 2016 — May 2020*

Experience

Graduate Research Assistant — Aug 2024 — Present
Iowa State University — *Ames, IA*

- Architected EM-based metagenomic classifier in Rust, leveraging FM-index for memory-efficient classification of paired-end Illumina reads
- Implemented compressed FM-index (suffix array → BWT → C array → Occ table) enabling terabyte-scale reference database handling
- Developed penalized Expectation-Maximization algorithm for accurate abundance estimation with convergence guarantees
- Ported core classification pipeline to WebGPU via wgpu, enabling browser-based inference on client-side hardware
- Authored Phylo-rs, a comprehensive Rust crate for phylogenetic analysis with 10x speedup over Python implementations
- Engineered WASM-compatible data structures for zero-copy deployment of bioinformatics algorithms to web clients

Research Intern — Aug 2023 — Aug 2024
ORISE, USDA-ARS — *Ames, IA*

- Developed AI infection identification models achieving high sensitivity, surpassing standard diagnostic benchmarks
- Designed Bayesian Deep Learning models for uncertainty quantification in Whole-Slide image segmentation
- Deployed an Active Learning algorithm that reduced labeling costs by optimizing sample selection for large-scale segmentation models

Undergraduate Research Intern — Nov 2019 — Nov 2020
Robert Bosch Center for Data Science and AI — *Chennai, India*

- Developed a novel Neural Network architecture for 3D brain tumor segmentation with improved accuracy
- Explored distributed inference strategies to enable deployment of large-scale models in resource-constrained hospital environments

Projects

Phylo-rs, Rust, WASM — Aug 2023 — Present

- Architected Phylo-rs, a Rust-based phylogenetic library outperforming existing Python tools by 10x in speed via memory-safe implementations
- Implemented advanced phylogenetic algorithms (SPR, diversity metrics) with WebAssembly support for cross-platform usage
- Built an extensible, memory-safe framework now serving as a foundation for safe phylogenetic inference tools

PREMISE, Rust, GPU, WebGPU — Jan 2026 — Present

- Engineered premise, an EM-based metagenomic classifier for paired-end Illumina reads with browser-based GUI
- Implemented compressed FM-index for memory-efficient large-scale reference databases
- Developed penalized Expectation-Maximization algorithm for accurate abundance estimation
- Ported full classification pipeline to WebGPU via wgpu, enabling client-side inference

webgpu-fmidx, Rust, WebGPU — Feb 2026 — Present

- Built GPU-accelerated FM-index construction for DNA sequences using compute shaders in WGSL
- Implemented radix sort, prefix sum, and gather shaders for parallel suffix array → BWT → C array → Occ table construction
- Achieved identical CPU/GPU results with zero-copy serialization via bincode
- Targeted both native (Vulkan/Metal/DX12) and WebAssembly deployments via wasm-pack

SmartHisto, Python, Computer Vision, PyTorch Aug 2021 — Dec 2023

- Engineered a high-throughput pipeline processing terabyte-scale whole-slide images, optimizing I/O bottlenecks for rapid training
- Developed an Active Learning framework to train Bayesian neural networks, significantly reducing annotation requirements

Selected Publications

- **Vijendran, S.**, Dorman, K., Anderson, T. K., & Eulenstein, O. "premise: an EM-based metagenomic classifier for paired-end Illumina reads." *bioRxiv* (2026).
- **Vijendran, S.**, Arruda, B., Anderson, T. K., & Eulenstein, O. "SmartHisto: Bayesian Active Learning for Histology Images." *bioRxiv* (2025).
- **Vijendran, S.**, Anderson, T., Markin, A., & Eulenstein, O. "Phylo-rs: an extensible phylogenetic analysis library in rust." *BMC Bioinformatics* 26, no. 1 (2025).
- Markin, A., **Vijendran, S.**, & Eulenstein, O. "Bounds on the Treewidth of Level-k Rooted Phylogenetic Networks." *arXiv* (2024).
- Nguyen, T.Q., Hutter, C.R., Markin, A., Thomas, M., Lantz, K., Killian, M.L., Janzen, G.M., **Vijendran, S.**, Wagle, S., Inderski, B., et al. "Emergence and interstate spread of highly pathogenic avian influenza A (H5N1) in dairy cattle in the United States." *Science* 388, no. 6745 (2025).
- Górecki, P., Markin, A., **Vijendran, S.**, & Eulenstein, O. "Computing generalized cophenetic distances under all Lp norms: A near-linear time algorithmic framework." *PLOS Computational Biology* 21, no. 6 (2025).
- **Vijendran, S.**, & Dubey, R. "Deep online sequential extreme learning machines and its application in pneumonia detection." *2019 8th International Conference on Industrial Technology and Management (ICITM)* (2019).

Technical Skills

- **Languages**: Rust, Python, C/C++, JavaScript, TypeScript, SQL, Bash, WGSL
- **ML & Data Science**: PyTorch, TensorFlow, Pandas, NumPy, Scikit-learn, Matplotlib, EM algorithms
- **Systems & Tools**: Git, Docker, WASM, WebGPU, wgpu, Linux, Vim, CI/CD, bincode
- **Web Frameworks**: Flask, Material-UI, HTML/CSS, Compute Shaders